

ProspectMap: The Two Ways a Mineral Prospectivity Map Lies, Quantified on the Real US Mississippi-Valley-Type Belt

Felipe Santibañez-Leal 

Abstract—A mineral prospectivity map stacks geophysical, geochemical and structural evidence over a study area and outputs, per cell, the probability that a deposit is there. The maps are routinely presented with a high area-under-curve (AUC) and a confident posterior, and both numbers are routinely inflated in two specific, well-understood ways. ProspectMap is an interactive tool that computes such a map by Weights of Evidence, a Bayesian log-odds update, and makes those two inflations *first-class*, measuring them rather than hiding them. This report quantifies both on the real US Midcontinent Mississippi-Valley-type zinc-lead belt (the open Critical Minerals Mapping Initiative dataset of Lawley *et al.*, 2022) and on ten synthetic terranes. The first inflation is *evaluation*: reporting AUC under random cross-validation, which lets spatially adjacent cells leak between train and test, overstates prospective skill; on the real belt the Weights-of-Evidence AUC falls from 0.72 under random cross-validation to 0.64 under spatial cross-validation (an inflation of 0.09, the largest of any case), and even a strong learned classifier drops from 0.98 to 0.95. The honest number is the spatial one, and on it the learned classifier (0.95) substantially beats classical Weights of Evidence (0.73). The second inflation is *modelling*: Weights of Evidence assumes the evidence layers are conditionally independent given a deposit, and it reports large contrasts that look decisive, but on the real belt those contrasts are statistically *insignificant*, every layer’s studentized contrast is below the 95% threshold, so the confident-looking weights do not actually survive their own error bars. The value of the tool is that it puts both truths on screen, the spatial-CV AUC and the studentized significance, rather than the flattering versions; every number and figure regenerates from the committed artifacts.

Index Terms—Mineral prospectivity, Weights of Evidence, conditional independence, spatial cross-validation, AUC inflation, Mississippi-Valley-type, critical minerals, honest evaluation, reproducibility.

I. INTRODUCTION

WHERE is the next ore deposit most likely to be? Mineral prospectivity mapping answers it by overlaying evidence layers, a magnetic high here, a geochemical anomaly there, proximity to a favourable structure, over a study-area grid and computing, per cell, a posterior probability of a deposit. The classical method is Weights of Evidence [2], [1]: each binary evidence pattern gets a positive weight W^+ where it is present and W^- where it is absent, their difference is the contrast $C = W^+ - W^-$, and the posterior log-odds of a deposit in a cell is the prior log-odds plus the sum of the

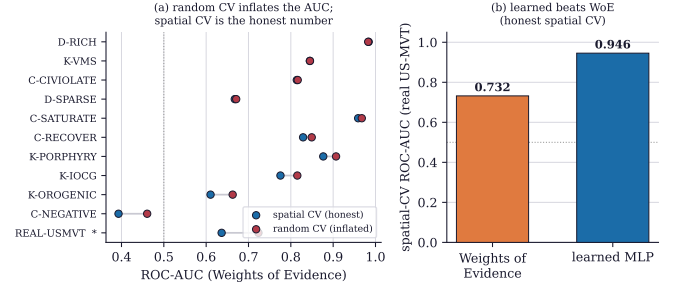


Fig. 1. The evaluation inflation, and the honest comparison. (a) Weights-of-Evidence AUC under random versus spatial cross-validation, per case: random cross-validation (red) is always at least as high as spatial (blue), because adjacent cells leak; the gap is largest on the real US-MVT belt (marked *), where the AUC falls from 0.72 to 0.64. Spatial cross-validation is the honest number. (b) On the real belt, evaluated under honest spatial cross-validation, a learned classifier (0.946) substantially outperforms classical Weights of Evidence (0.732).

present-or-absent weights. The output is a map, and the map is usually sold with two numbers: a high area-under-curve showing it separates known deposits from barren ground, and confident-looking weights. Both are, in a well-documented way, prone to inflation.

ProspectMap is a tool built to expose the inflation rather than enjoy it. It computes the Weights-of-Evidence posterior live, and it makes first-class the two ways the numbers lie. The first is that an AUC reported under *random* cross-validation is optimistic, because a prospectivity grid is spatially autocorrelated and a random split leaves near-neighbour cells in both train and test, so the model is partly tested on what it has seen; *spatial* cross-validation, holding out contiguous blocks, gives the honest number [5]. The second is that Weights of Evidence assumes the evidence layers are conditionally independent given a deposit, and when they are not the posterior is overconfident [3]; and even when the contrasts look large, they may not be statistically significant. This report quantifies both, on the real US Mississippi-Valley-type zinc-lead belt from the open Critical Minerals Mapping Initiative dataset [4] and on ten synthetic terranes, and shows that a modern learned classifier, evaluated honestly, does better than the classical method [6]. The contribution is the measurement and the display of the two inflations, not a new prospectivity algorithm.

F. Santibañez-Leal is with the CAOS open-research programme, Santiago, Chile (e-mail: fsantibanez@gmail.com; ORCID 0000-0002-0150-3246).

Technical report, 2026. Interactive tool and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ProspectMap; live at <https://prospectmap.fasl-work.com>.

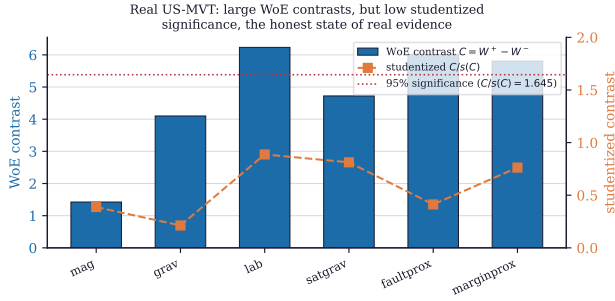


Fig. 2. The modelling inflation. Weights-of-Evidence contrasts $C = W^+ - W^-$ (bars) and their studentized significance $C/s(C)$ (line) for the six evidence layers of the real US-MVT belt. The raw contrasts are large (up to 6), which looks decisive, but every layer’s studentized contrast is below the 95% significance threshold of 1.645: on real, sparse deposit data the weights carry large standard errors and are not statistically significant, so the confident-looking map rests on evidence that does not survive its own error bars.

II. THE FIRST LIE: CROSS-VALIDATION INFLATION

The first inflation is a matter of how the map is scored, and it is quantified in Fig. 1a. For every case, the Weights-of-Evidence posterior is evaluated two ways: random five-fold cross-validation, which splits the grid cells at random, and spatial cross-validation, which holds out contiguous blocks of cells so that the test set is geographically separate from the training set. Random cross-validation is always at least as optimistic, because a prospectivity grid is strongly spatially autocorrelated and a random split leaves a test cell’s near neighbours in the training set, so the model is scored partly on cells like ones it has seen. The gap is real and, on the *real* US Mississippi-Valley-type belt, it is the largest of any case: the Weights-of-Evidence AUC falls from 0.723 under random cross-validation to 0.637 under spatial, an inflation of 0.087, and a strong learned classifier falls from 0.982 to 0.946. The honest number is the spatial one, and reporting the random one, as much of the literature does, overstates how well the map would find a genuinely unseen deposit. On that honest spatial-CV footing (Fig. 1b) the comparison of methods is clear: a learned classifier reaches 0.946 against classical Weights of Evidence’s 0.732, because the learned model captures interactions between evidence layers that the conditional-independence assumption of Weights of Evidence discards.

III. THE SECOND LIE: OVERCONFIDENT WEIGHTS

The second inflation is in the weights themselves (Fig. 2). Weights of Evidence assumes the evidence layers are conditionally independent given a deposit; when correlated layers are stacked, their weights double-count the same signal and the posterior becomes overconfident, which is why a conditional-independence test [3] belongs in any honest workflow, and ProspectMap runs one. But there is a simpler and equally important check that is often skipped: whether a contrast is statistically significant at all. On the real US-MVT belt the six evidence layers have large raw contrasts, up to about 6, which a practitioner reading only the contrast would take as strong evidence. Their studentized contrasts $C/s(C)$, however, are all

between 0.2 and 0.9, well below the 1.645 threshold for 95% significance: with sparse deposits and noisy layers the standard error of each weight is large, and none of the contrasts actually clears its own error bar. The map built from them may still be useful for ranking, but presenting its weights as decisive evidence is the second way the map lies, and showing the studentized contrast beside the raw one is the correction.

IV. DISCUSSION AND SCOPE

The two inflations are independent and both matter. A map can have honest, significant weights and still report an inflated AUC because it was scored with random cross-validation; or it can have an honestly-scored AUC and still oversell weights that are not significant. ProspectMap’s contribution is to compute the Weights-of-Evidence map and then put the deflating diagnostics next to it, the spatial-CV AUC beside the random one, the studentized contrast beside the raw one, and a conditional-independence test, so a user sees the honest picture rather than the flattering one, and the comparison with a learned classifier shows what honest evaluation is worth (the learned model’s advantage over Weights of Evidence, 0.946 against 0.732, is real precisely because it is measured under spatial cross-validation). The scope is stated plainly. The ten terrane cases are synthetic, used to exercise the diagnostics under known ground truth; the one real case is the open Critical Minerals Mapping Initiative US-MVT dataset [4], a genuine belt, and its honest numbers are the sobering ones a real prospectivity study should expect. The tool is a decision-support and teaching instrument, not an exploration recommendation for any specific ground; it says where the evidence points and, more importantly, how much to trust it. None of this is hidden; the interface shows the same diagnostics on every map.

V. CONCLUSION

A mineral prospectivity map lies in two well-understood ways: it reports an AUC inflated by random cross-validation, and it presents Weights-of-Evidence contrasts that look decisive but may violate conditional independence or fail their own significance test. ProspectMap measures both. On the real US Mississippi-Valley-type belt the Weights-of-Evidence AUC falls from 0.72 to 0.64 when scored honestly by spatial cross-validation (the largest inflation of any case), a learned classifier evaluated the same honest way reaches 0.95 against Weights of Evidence’s 0.73, and every evidence layer’s contrast is statistically insignificant despite looking large. Putting the honest numbers on screen next to the flattering ones, spatial-CV AUC and studentized significance, is the report’s discipline, and the reproducible tool that does it is its contribution.

APPENDIX A REPRODUCTION

The committed artifacts under `data/derived/` carry each case’s Weights-of-Evidence layers (contrasts and studentized contrasts), posterior summary, pairwise conditional-independence tests, and both random- and spatial-cross-validation AUC (`case-results.json`),

together with the learned-classifier comparison on the real belt (pm-learned-real.json) and the real US-MVT trace (REAL-USMVT, with the Lawley *et al.* provenance). The figure scripts under manuscripts/prospectivity/figures/ read those artifacts; the pipeline is seeded, so re-running reproduces Fig. 1 and Fig. 2.

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